## ASCII MASTER FLOW — PANEL 1/12: Savanna transition map

```
EQUATORWARD -> wetter and taller grass
SUDAN BELT -> classic wet-dry savanna
POLEWARD -> shorter grass and open trees
DESERT MARGIN -> drought risk intensifies
MUST REMEMBER: Savanna Aw/As climates reflect seasonal ITCZ migration between convective wet
  summers and subtropical-high dry winters, sustaining tall grasses with scattered
  drought/fire-adapted trees across Africa, South America and northern Australia.
```

## ASCII MASTER FLOW — PANEL 2/12: ITCZ seasonal switch

```
HIGH-SUN SEASON -> ITCZ approaches
CONVERGENCE + CONVECTION -> summer rain
LOW-SUN SEASON -> dry trades dominate
RESULT -> hot wet / warm dry rhythm
```

## ASCII MASTER FLOW — PANEL 3/12: Water-reliability chain

```
LATE ONSET -> delayed sowing
MID-SEASON BREAK -> crop and forage stress
EARLY CESSATION -> incomplete grain filling
DROUGHT -> migration and herd loss risk
```

## ASCII MASTER FLOW — PANEL 4/12: Grass-tree gradient

```
WET MARGIN -> tall grass + more trees
CORE SAVANNA -> parkland mosaic
DRY MARGIN -> short grass + thorny shrubs
DESERT EDGE -> discontinuous vegetation
```

## ASCII MASTER FLOW — PANEL 5/12: Fire-herbivory balance

```
SEASONAL RAIN -> grass biomass
DRY SEASON -> combustible fuel
FIRE + BROWSING -> seedling suppression
THICK BARK + ROOTS -> selected tree survival
```

## ASCII MASTER FLOW — PANEL 6/12: Wildlife-livelihood matrix

```
HERDS -> follow seasonal forage and water
PREDATORS -> track herbivore concentrations
PASTORALISTS -> mobility spreads risk
CULTIVATORS -> millet / sorghum timing
```

## ASCII MASTER FLOW — PANEL 7/12: Degradation pathway

COVER LOSS -> raindrop and wind exposure
WET SEASON -> runoff, leaching and erosion
DRY SEASON -> fire and surface sealing
FIXED WATER POINT -> local grazing concentration

## ASCII MASTER FLOW — PANEL 8/12: India forest gradient

```
MOIST DECIDUOUS -> longer wet season
DRY DECIDUOUS -> stronger leaf-shedding
THORN / SCRUB -> semi-arid moisture stress
RULE -> overlapping belts, not fixed walls
```

## ASCII MASTER FLOW — PANEL 9/12: Indian grassland firewall

```
TERAI -> humid tall grass and floodplain
SHOLA -> montane grass-forest mosaic
BANNI -> arid-saline Kachchh grassland
MANAGEMENT -> hydrology and history first
```

## ASCII MASTER FLOW — PANEL 10/12: Restoration decision tree

```
NATURAL OPEN ECOSYSTEM -> conserve openness
DEGRADED SCRUB -> restore native mosaic
WATER PROCESS -> recharge and drainage
LIVELIHOOD -> grazing and tenure design
CLOSE DISTINCTION: Savanna is tropical grassland, steppe is semi-arid temperate/subtropical
  grassland, and parkland structure is not evidence of a climax maintained by climate alone
  because fire, herbivory and land use matter.
```

## ASCII MASTER FLOW — PANEL 11/12: Savanna PYQ firewall

```
2021 PRELIMS -> tree-limiting conditions
CLIMATE -> long dry season
DISTURBANCE -> fire + herbivory
LEDGER -> no answer letter invented
EVIDENCE LIMIT: Fire can recycle nutrients and maintain openness but frequency/intensity
  changes may degrade systems; Indian deciduous forests and grasslands need region-specific
  rainfall, fire and grazing evidence rather than direct Sudan-type equivalence.
```

## ASCII MASTER FLOW — PANEL 12/12: Savanna answer spine

```
LOCATE -> rainforest-desert transition
DERIVE -> ITCZ and wet-dry season
EXPLAIN -> grass-tree disturbance balance
APPLY -> India mosaics and restoration
```